

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **5**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 5****System A**

$$\begin{cases} -2x - 4y = 10 & \text{[A1]} \\ -6x - 5y = 9 & \text{[A2]} \end{cases}$$

System B

$$\begin{cases} -2x - 4y = 10 & \text{[B1]} \\ 7y = -21 & \text{[B2]} \end{cases}$$

System C

$$\begin{cases} -2x - 4y = 10 & \text{[C1]} \\ y = -3 & \text{[C2]} \end{cases}$$

(a) How do we transform System A into System B?

- ☐ ___ $\times$ Equation [A1] $\rightarrow$ Equation [B1]
- ☐ ___ $\times$ Equation [A2] $\rightarrow$ Equation [B2]
- ☐ ___ $\times$ Equation [A1] + Equation [A2] $\rightarrow$ Equation [B2]
- ☐ ___ $\times$ Equation [A2] + Equation [A1] $\rightarrow$ Equation [B1]

(b) How do we transform System B into System C?

- ☐ ___ $\times$ Equation [B1] $\rightarrow$ Equation [C1]
- ☐ ___ $\times$ Equation [B2] $\rightarrow$ Equation [C2]
- ☐ ___ $\times$ Equation [B1] + Equation [B2] $\rightarrow$ Equation [C2]
- ☐ ___ $\times$ Equation [B2] + Equation [B1] $\rightarrow$ Equation [C1]

Question 2 of 5

Consider the following system of linear equations.

$$4x - 16y = -4$$

$$-4x + 21y = -11$$

Solve the system by completing the steps below to produce a reduced row-echelon form. R_1 and R_2 denote the first and second rows, respectively. The arrow notation ($\rightarrow$) means the expression/matrix on the left becomes the expression/matrix on the right once the row operations are performed.

(a) For each step below, enter the coefficient for the row operation. The first matrix in Step 1 is the augmented matrix for the given system of equations.

Step 1:

$$\left[\begin{array}{cc|c} 4 & -16 & -4 \\ -4 & 21 & -11 \end{array} \right] \xrightarrow{\text{---} \cdot R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ -4 & 21 & -11 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ -4 & 21 & -11 \end{array} \right] \xrightarrow{\text{---} \cdot R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 5 & -15 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 5 & -15 \end{array} \right] \xrightarrow{\text{---} \cdot R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 1 & -3 \end{array} \right]$$

Step 4: Enter the coefficient for the row operations and the missing entries in the resulting matrix.

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 1 & -3 \end{array} \right] \xrightarrow{\text{---} \cdot R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & \text{---} \\ 0 & 1 & \text{---} \end{array} \right]$$

(b) Give the solution.

$x = \text{---}$ $y = \text{---}$

Question 3 of 5

Consider the following system of linear equations.

$$-5x + 10y = -25$$

$$6x - 10y = 25$$

Solve the system by completing the steps below to produce a reduced row-echelon form. R_1 and R_2 denote the first and second rows, respectively. The arrow notation ($\rightarrow$) means the expression/matrix on the left becomes the expression/matrix on the right once the row operations are performed.

(a) Enter the augmented matrix.

$$\left[\begin{array}{cc|c} _ & _ & _ \\ _ & _ & _ \end{array} \right]$$

(b) For each step below, enter the coefficient for the row operation and the missing entries in the resulting matrix.

Step 1:

$$\boxed{} \cdot R_1 \rightarrow R_1$$

$$\left[\begin{array}{cc|c} 1 & _ & _ \\ _ & _ & _ \end{array} \right]$$

Step 2:

$$\boxed{} \cdot R_1 + R_2 \rightarrow R_2$$

$$\left[\begin{array}{cc|c} 1 & _ & _ \\ 0 & _ & _ \end{array} \right]$$

Step 3:

$$\boxed{} \cdot R_2 \rightarrow R_2$$

$$\left[\begin{array}{cc|c} 1 & _ & _ \\ 0 & 1 & _ \end{array} \right]$$

Step 4:

$$\boxed{} \cdot R_2 + R_1 \rightarrow R_1$$

$$\left[\begin{array}{cc|c} 1 & 0 & _ \\ 0 & 1 & _ \end{array} \right]$$

(c) Give the solution.

$$x = \boxed{} \quad y = \boxed{}$$

Question 4 of 5

Consider the following system of linear equations.

$$\begin{cases} 5x & + 10z = -50 \\ 4x - 2y & = -30 \\ 2x - 3y - 4z & = -17 \end{cases}$$

Solve the system by completing the steps below to produce a reduced row-echelon form. R_1 , R_2 , and R_3 denote the first, second, and third rows, respectively. The arrow notation ($\rightarrow$) means the expression/matrix on the left becomes the expression/matrix on the right once the row operations are performed.

(a) For each step below, enter the coefficient for the row operation. (The first matrix in Step 1 is the augmented matrix for the given system of equations.) For Step 6, also fill in the missing matrix entries.

Step 1:

$$\left[\begin{array}{ccc|c} 5 & 0 & 10 & -50 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right] \xrightarrow{\boxed{} \cdot R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right] \xrightarrow{\begin{array}{l} -4 \cdot R_1 + R_2 \rightarrow R_2 \\ \boxed{} \cdot R_1 + R_3 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & -2 & -8 & 10 \\ 0 & -3 & -8 & 3 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & -2 & -8 & 10 \\ 0 & -3 & -8 & 3 \end{array} \right] \xrightarrow{\boxed{} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & -3 & -8 & 3 \end{array} \right]$$

Step 4:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & -3 & -8 & 3 \end{array} \right] \xrightarrow{\boxed{} \cdot R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 4 & -12 \end{array} \right]$$

Step 5:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 4 & -12 \end{array} \right] \xrightarrow{\boxed{} \cdot R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

Step 6:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 1 & -3 \end{array} \right] \xrightarrow{\begin{array}{l} -2 \cdot R_3 + R_1 \rightarrow R_1 \\ \boxed{} \cdot R_3 + R_2 \rightarrow R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \boxed{} \\ 0 & 1 & 0 & \boxed{} \\ 0 & 0 & 1 & \boxed{} \end{array} \right]$$

(b) Give the solution.

$x = \boxed{}$

$y = \boxed{}$

$z = \boxed{}$

Question 5 of 5

Suppose that the augmented matrix of a system of linear equations for unknowns x , y , and z is

$$\left[\begin{array}{ccc|c} 1 & \frac{21}{4} & 4 & \frac{21}{4} \\ 2 & \frac{21}{2} & 8 & \frac{21}{2} \\ 5 & \frac{105}{4} & 20 & \frac{105}{4} \end{array} \right].$$

Solve the system and provide the information requested.

The system has:

___ a unique solution, which is

$x = \underline{\hspace{1cm}} \quad y = \underline{\hspace{1cm}} \quad z = \underline{\hspace{1cm}}$

___ infinitely many solutions, two of which are

$x = \underline{\hspace{1cm}} \quad y = \underline{\hspace{1cm}} \quad z = \underline{\hspace{1cm}}$

$x = \underline{\hspace{1cm}} \quad y = \underline{\hspace{1cm}} \quad z = \underline{\hspace{1cm}}$

___ no solution

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Question 1 of 5

(a) How do we transform System A into System B?

- ☐ $\square \times \text{Equation [A1]} \rightarrow \text{Equation [B1]}$
- ☐ $\square \times \text{Equation [A2]} \rightarrow \text{Equation [B2]}$
- ☒ $-3 \times \text{Equation [A1]} + \text{Equation [A2]} \rightarrow \text{Equation [B2]}$
- ☐ $\square \times \text{Equation [A2]} + \text{Equation [A1]} \rightarrow \text{Equation [B1]}$

(b) How do we transform System B into System C?

- ☐ $\square \times \text{Equation [B1]} \rightarrow \text{Equation [C1]}$
- ☒ $\frac{1}{7} \times \text{Equation [B2]} \rightarrow \text{Equation [C2]}$
- ☐ $\square \times \text{Equation [B1]} + \text{Equation [B2]} \rightarrow \text{Equation [C2]}$
- ☐ $\square \times \text{Equation [B2]} + \text{Equation [B1]} \rightarrow \text{Equation [C1]}$

Question 2 of 5

(a) For each step below, enter the coefficient for the row operation. The first matrix in Step 1 is the augmented matrix for the given system of equations.

Step 1:

$$\left[\begin{array}{cc|c} 4 & -16 & -4 \\ -4 & 21 & -11 \end{array} \right] \xrightarrow{\frac{1}{4} \cdot R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ -4 & 21 & -11 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ -4 & 21 & -11 \end{array} \right] \xrightarrow{4 \cdot R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 5 & -15 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 5 & -15 \end{array} \right] \xrightarrow{\frac{1}{5} \cdot R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 1 & -3 \end{array} \right]$$

Step 4: Enter the coefficient for the row operations and the missing entries in the resulting matrix.

$$\left[\begin{array}{cc|c} 1 & -4 & -1 \\ 0 & 1 & -3 \end{array} \right] \xrightarrow{4 \cdot R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & -13 \\ 0 & 1 & -3 \end{array} \right]$$

(b) Give the solution.

$$x = -13$$

$$y = -3$$

Question 3 of 5

(a) Enter the augmented matrix.

$$\left[\begin{array}{cc|c} -5 & 10 & -25 \\ 6 & -10 & 25 \end{array} \right]$$

(b) For each step below, enter the coefficient for the row operation and the missing entries in the resulting matrix.

Step 1:

$$-\frac{1}{5} \cdot R_1 \rightarrow R_1$$

$$\left[\begin{array}{cc|c} 1 & -2 & 5 \\ 6 & -10 & 25 \end{array} \right]$$

Step 2:

$$-6 \cdot R_1 + R_2 \rightarrow R_2$$

$$\left[\begin{array}{cc|c} 1 & -2 & 5 \\ 0 & 2 & -5 \end{array} \right]$$

Step 3:

$$\frac{1}{2} \cdot R_2 \rightarrow R_2$$

$$\left[\begin{array}{cc|c} 1 & -2 & 5 \\ 0 & 1 & -\frac{5}{2} \end{array} \right]$$

Step 4:

$$2 \cdot R_2 + R_1 \rightarrow R_1$$

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -\frac{5}{2} \end{array} \right]$$

(c) Give the solution.

$$x = 0 \quad y = -\frac{5}{2}$$

Question 4 of 5

(a) For each step below, enter the coefficient for the row operation. (The first matrix in Step 1 is the augmented matrix for the given system of equations.) For Step 6, also fill in the missing matrix entries.

Step 1:

$$\left[\begin{array}{ccc|c} 5 & 0 & 10 & -50 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right] \xrightarrow{\frac{1}{5} \cdot R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 4 & -2 & 0 & -30 \\ 2 & -3 & -4 & -17 \end{array} \right] \xrightarrow{\begin{array}{l} -4 \cdot R_1 + R_2 \rightarrow R_2 \\ -2 \cdot R_1 + R_3 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & -2 & -8 & 10 \\ 0 & -3 & -8 & 3 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & -2 & -8 & 10 \\ 0 & -3 & -8 & 3 \end{array} \right] \xrightarrow{\frac{-1}{2} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & -3 & -8 & 3 \end{array} \right]$$

Step 4:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & -3 & -8 & 3 \end{array} \right] \xrightarrow{3 \cdot R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 4 & -12 \end{array} \right]$$

Step 5:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 4 & -12 \end{array} \right] \xrightarrow{\frac{1}{4} \cdot R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

Step 6:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -10 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 1 & -3 \end{array} \right] \xrightarrow{\begin{array}{l} -2 \cdot R_3 + R_1 \rightarrow R_1 \\ -4 \cdot R_3 + R_2 \rightarrow R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

(b) Give the solution.

$$x = -4$$

$$y = 7$$

$$z = -3$$

Question 5 of 5

The system has:

___ infinitely many solutions, two of which are

$$x = \frac{21}{4} - \frac{21}{4}, y = 1, z = 0$$

$$x = \frac{21}{4} - 4, y = 0, z = 1$$